

CAR RENTAL REQUEST AUTOMATION IN SERVICENOW

Phase 5: Project Development Phase

Records how the solution was built and tested, with screenshots from the working ServiceNow instance.

Item	Details
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Platform	ServiceNow (Service Catalog, Flow Designer, Business Rules)
Document	Phase 5 of 5

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5.1 Development Overview and Build Order

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5.2 Environment Setup

- A ServiceNow developer instance with administrative access.

- Service Catalog and Flow Designer (Workflow Studio) available.
- A named Update Set selected as the current set before starting work.

5.3 Service Catalog Configuration

Activity 1: Create the catalog item

1. Open ServiceNow, click **All**, and search for **Maintain Items**.
2. Click **New** and fill in the details: name **Car Rental Request**, catalog **Service Catalog**, category **Transportation Services**, short description **Request a car for rental**.
3. Click **Save**. The item is Active and its State is Published.

The screenshot shows the ServiceNow interface for creating a new catalog item. The browser address bar displays the URL: `https://dev427451.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3D3D390c9f72f53831003fff0807fa4e31f...`. The page title is "Catalog Item - Car Rental Request". The "Application" is set to "Global". The "Active" checkbox is checked. The "Fulfillment automation level" is set to "Unspecified". The "Name" field is "Car Rental Request". The "Catalogs" field is "Service Catalog". The "Category" is "Transportation Services". The "State" is "Published". The "Checked out" field is "false". The "Owner" is "System Administrator". The "Secondary owners" field is empty. The "Short description" is "Request a car for rental". The "Description" field is empty. The "Item Details" tab is selected, and the "Process Engine" tab is also visible. The "Pricing" and "Portal Settings" tabs are also visible. The "Description" field has a rich text editor with a toolbar showing bold, italic, underline, link, unlink, and other formatting options.

Figure 5.1: Catalog item record for Car Rental Request

Activity 2: Create the catalog variables

After saving, open the **Variables** related list and create the seven variables. The list below is the final result.

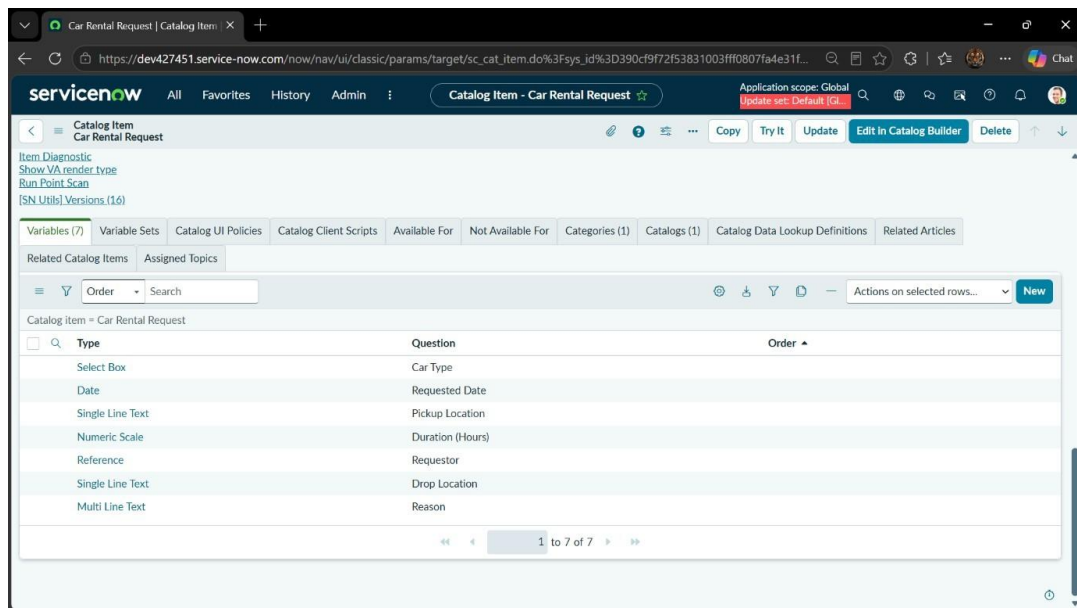


Figure 5.2: Seven catalog variables on the Car Rental Request item

For the **Car Type** variable, open the Question Choices tab and add three choices: **Mini**, **SUV**, and **Sedan**. Save the catalog item again. The variables and choices appear on the requester's form exactly as shown in Figure 3.3 of the Design document.

5.4 Flow Designer Implementation

Activity 1: Create the flow and trigger

1. Open **All**, search for **Flow Designer**, then click **New** and choose **Flow**.
2. Name the flow **Car Rental Req**. Open **Show additional properties** and under **Run** select **System User**.
3. Click **Build Flow**, then click **+** to add a trigger, select **Service Catalog**, and click **Done**.

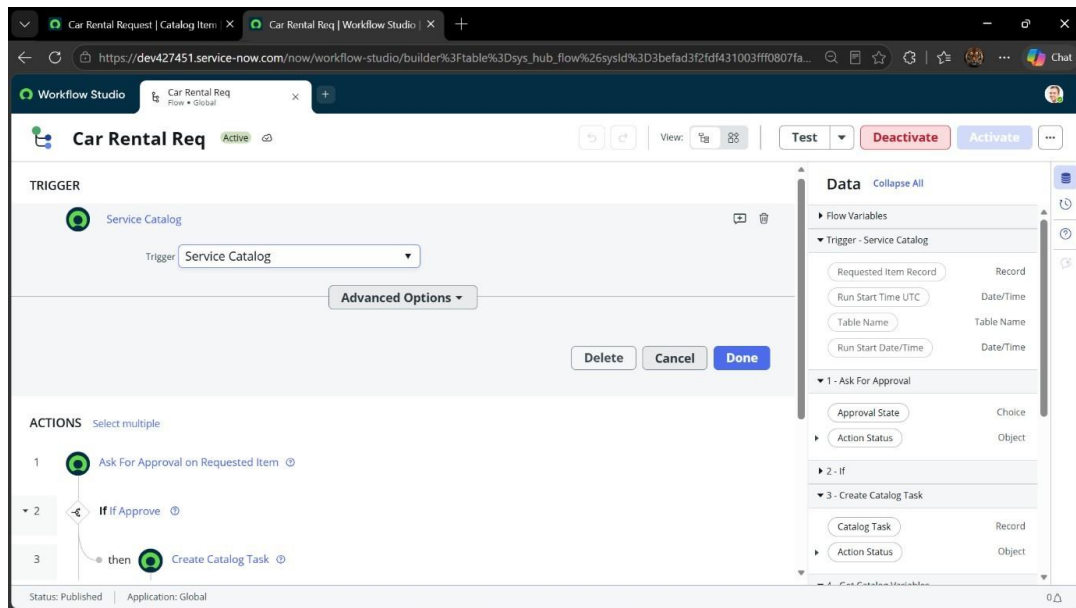


Figure 5.3: Service Catalog trigger of the Car Rental Req flow

Activity 2: Ask For Approval

1. Click **Add an Action** and choose **Ask For Approval**.
2. Drag the **Requested Item Record** from the trigger into the Record field. The table is Requested Item (sc_req_item).
3. Set **Approval Field** to **Approval** and **Journal Field** to **Approval history**.
4. Under Rules, choose **Approve** when **Anyone approves**, and select the approver.

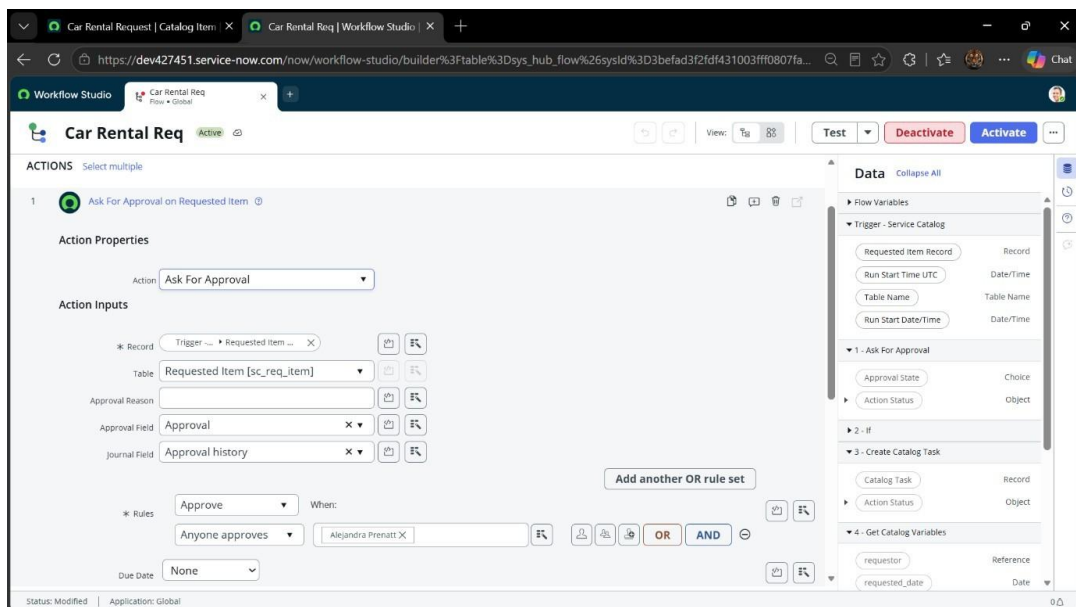


Figure 5.4: Ask For Approval action on the Requested Item

Concept note: how Ask For Approval works. The action sets the approval field on the RITM to *Requested*, creates approval records for the approver, and **pauses the flow**. When the approver decides, the field is updated and the flow **resumes**. The next step reads the outcome through the *Approval State* output pill.

Activity 3: Add the condition

1. Add an **If** flow logic step and set the Condition Label to **If Approve**.
2. Using the data pill picker, choose **Ask For Approval > Approval State**, the operator **is**, and the value **Approved**.
3. Click **Done**. Everything nested under this step runs only for approved requests.

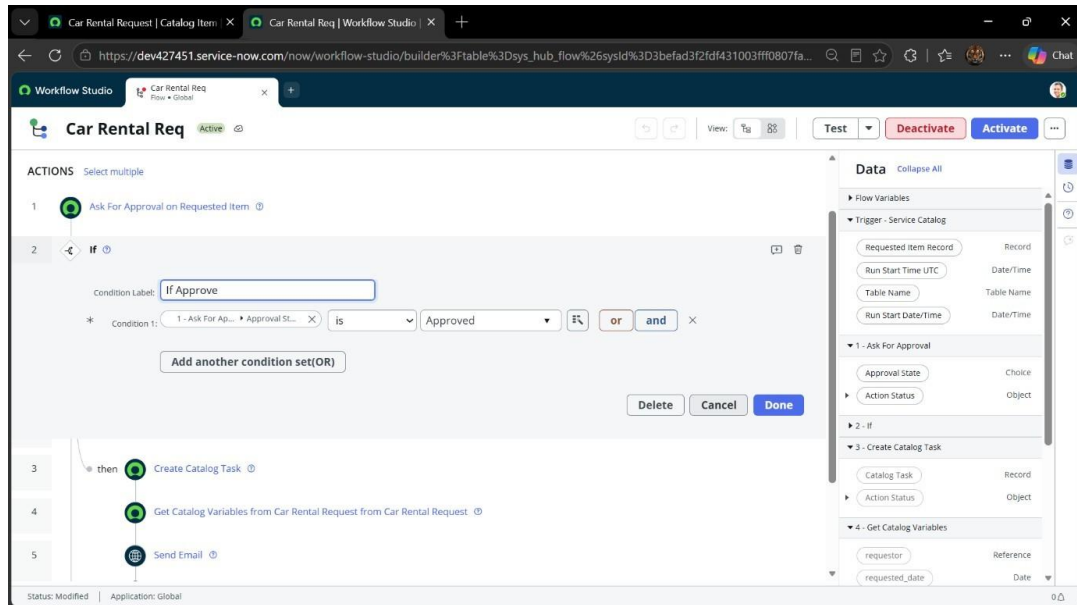


Figure 5.5: If condition checking that Approval State is Approved

Activity 4: Create the Catalog Task

1. Inside the **If** branch, add the action **Create Catalog Task**.
2. Set **Table Name** to **Catalog Task [sc_task]** and drag the Requested Item from the trigger into the Requested Item field.
3. Add two fields: **Assignment group = Transport Team** and **Description** ("Verify This Item And Assign To A Person Who Manages").
4. Select the template catalog item **Car Rental Request**. The **Wait** option is enabled.

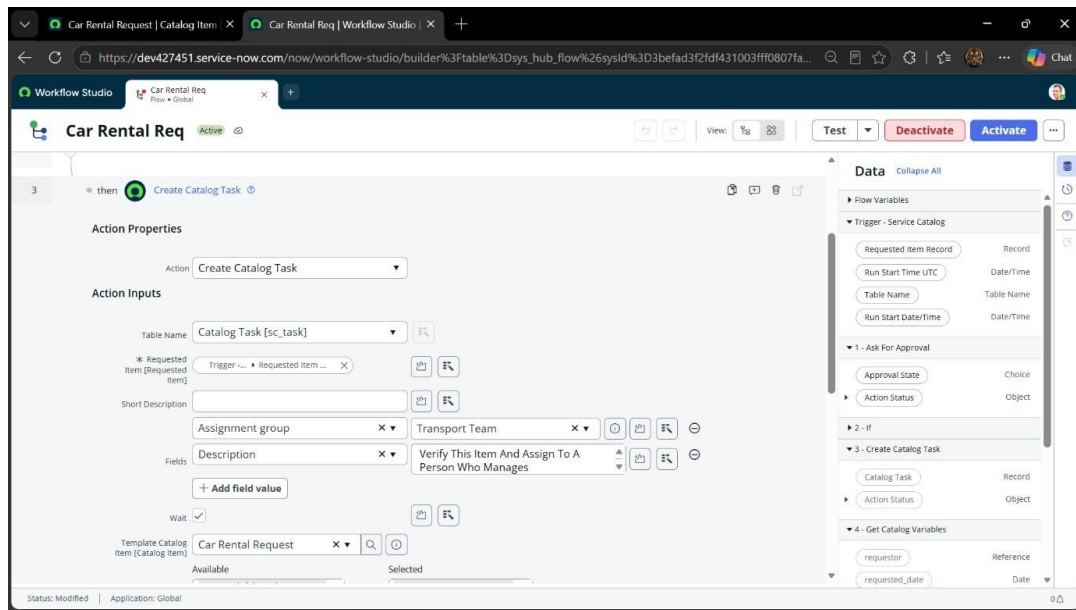


Figure 5.6: Create Catalog Task assigned to the Transport Team

Concept note: the Wait option. With **Wait** enabled, the flow does not move to the next action as soon as the task is created. It pauses until the catalog task is completed. This means the email in Activity 6 is sent **after the Transport team closes the task**, not at the moment of approval.

Activity 5: Get Catalog Variables

1. Add the action **Get Catalog Variables**.
2. Set **Submitted Request** to the Requested Item from the trigger and the template to **Car Rental Request**.
3. Move the variables from Available to **Selected** (requestor, requested_date, pickup_location, drop_location, duration_hrs, car_type, and so on). Click **Done**.

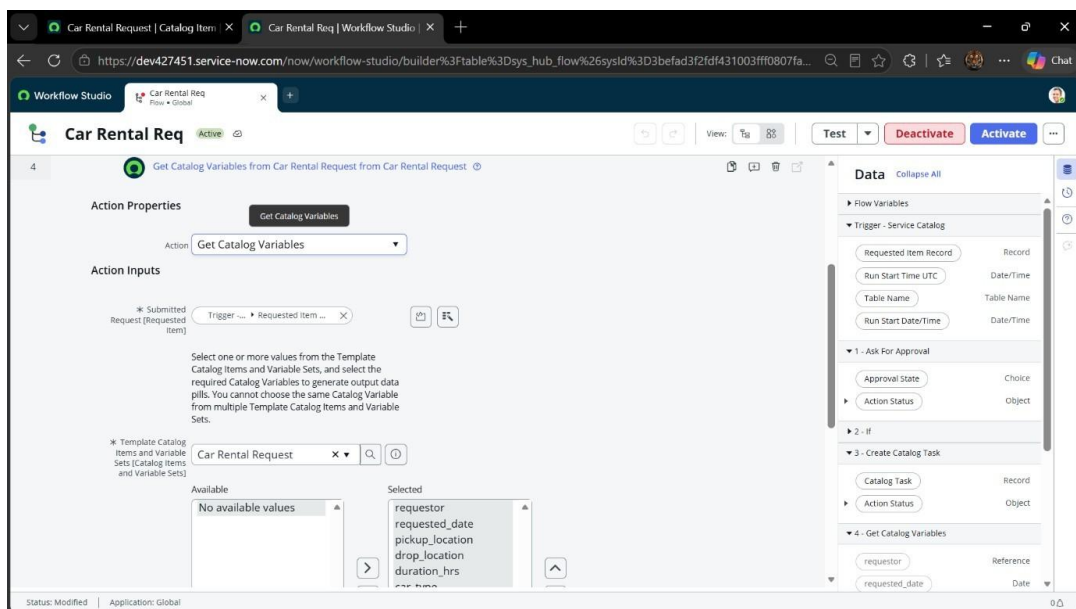


Figure 5.7: Get Catalog Variables with the selected variables

Activity 6: Send Email

1. Add the action **Send Email** and drag the Requested Item as the **Target Record** (table `sc_req_item`).
2. In **To**, use the data pill picker: **Get Catalog Variables > requestor > Email**.
3. Set **Subject** to `Your Car Request is approved.`
4. In the **Body**, enter: "Dear, Your car Rental Request is approved! Regards." Click **Done**.

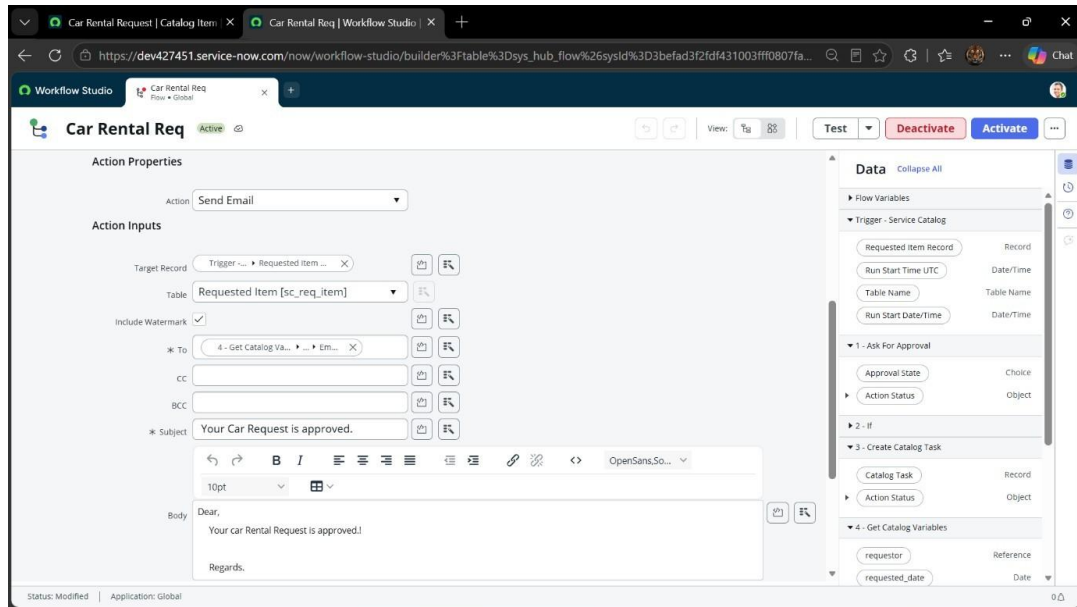


Figure 5.8: Send Email action addressed to the requestor

Finally, **Save** and **Activate** the flow. The flow status shows **Active** and **Published**, as seen in Figure 3.4 of the Design document.

5.5 Business Rule: Create Incident If Car Unavailable

A Business Rule handles the exception case. It is defined on the **Requested Item [sc_req_item]** table, runs **after** insert and update, and has an order of 100. It is an advanced rule, so it uses a script.

The screenshot shows the ServiceNow interface for configuring a Business Rule. The rule is named 'Create Incident If Car Unavailable' and is applied to the 'Requested Item [sc_req_item]' table. It is set to run 'after' insert and update events, with an order of 100. The rule is marked as 'Active' and 'Advanced'. The 'When to run' tab is selected, showing the trigger configuration.

Figure 5.9: Business Rule settings (table, timing, and triggers)

When the record's state equals 3, the script creates an incident whose short description names the request number and whose description includes the requested date.

The screenshot shows the ServiceNow interface for editing the script of the Business Rule. The script is written in JavaScript and is designed to create an incident when the state of the requested item is 3. The script uses the `GLideRecord` class to create and insert the incident record.

```

1  (function executeRule(current, previous /*null when async*/) {
2
3
4      if (current.state == 3) {
5
6          var inc = new GlideRecord('incident');
7
8          inc.initialize();
9
10         inc.short_description = 'Car unavailable for request: ' + current.number;
11
12         inc.description = 'Car not available for trip on ' + current.variables.requested_date;
13
14         inc.insert();
15     }
16 }
17
18 })(current, previous);

```

Figure 5.10: Business Rule script that creates the incident

The script, as configured:

```
(function executeRule(current, previous /*null when async*/ ) {

    if (current.state == 3) {

        var inc = new GlideRecord('incident');
        inc.initialize();
        inc.short_description = 'Car unavailable for request: ' + current.number;
        inc.description = 'Car not available for trip on ' +
            current.variables.requested_date;
        inc.insert();

    }

})(current, previous);
```

5.6 Attach the Flow to the Catalog Item

1. Open the Car Rental Request catalog item record.
2. On the **Process Engine** tab, select the Car Rental Req flow.
3. Save the record.

Once linked, every submitted request follows the predefined approval and processing steps. The flow reads the form data, triggers the approval, creates the task, and sends the notification, which gives consistency, accuracy, and traceability.

5.7 Configuration Summary (As Built)

Object	Setting	Value
Catalog item	Name / catalog / category	Car Rental Request / Service Catalog / Transportation Services
Catalog item	State / owner	Published / System Administrator
Variables	Count	7 (Car Type, Requested Date, Pickup Location, Duration (Hours), Requestor, Drop Location, Reason)
Flow	Name / status	Car Rental Req / Active, Published
Flow	Trigger	Service Catalog
Ask For Approval	Field / journal / rule	Approval / Approval history / Approve when Anyone approves
If	Condition	Approval State is Approved
Create Catalog Task	Table / group / Wait	sc_task / Transport Team / enabled
Send Email	To / subject	Requestor's email / Your Car Request is approved.
Business Rule	Table / when / order	sc_req_item / after insert and update / 100

5.8 Testing and Results

Functional tests

#	Test	Steps	Expected result
T1	Item visible	Open the catalog and find Car Rental Request	Item appears under Transportation Services with all seven fields
T2	Car Type choices	Open the Car Type dropdown	Mini, SUV, and Sedan are listed
T3	Submit request	Fill in the form and click Order Now	Confirmation shows a Request Number (REQ . . .)
T4	Records created	Open the REQ, then the RITM	RITM is linked to the REQ and the variables are saved
T5	Approval generated	Open the RITM and check Approvers	An approval record is in the Requested state
T6	Approve	The approver clicks Approve	Approval state on the RITM becomes Approved
T7	Task creation	Check catalog tasks after approval	An <code>sc_task</code> exists with assignment group Transport Team
T8	Email	Close the catalog task, then check the email log	"Your Car Request is approved." is sent to the requestor
T9	Flow execution	Flow Designer, Executions	Every action shows Completed

Negative and exception tests

#	Test	Steps	Expected result
T10	Rejection	The approver rejects the request	Approval State is not Approved: no task and no email are created
T11	Unavailable car	Set the RITM state to the value the rule checks (3)	An incident is created with the short description "Car unavailable for request: RITM..."

Evidence from the working instance for the steps above:

The screenshot shows the ServiceNow 'Order Status' page for request REQ0010004. The page header includes the ServiceNow logo, navigation links, and the request ID. A green notification bar at the top states 'Thank you, your request has been submitted'. Below this, the order details are listed: Order Placed: 2026-09-20 04:41:15, Request Number: REQ0010004, and Estimated Delivery Date of Complete Order: 2026-09-20. A table displays the order items:

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
Request a car for rental	2026-09-20			1	
Total					-

Buttons for 'Back to Catalog', 'Continue Shopping', and 'Home' are visible at the bottom of the table.

Order confirmation page showing the Request Number

The screenshot shows the ServiceNow 'Requested Item' record for RITM0010010. The page header includes the ServiceNow logo, navigation links, and the request ID. A dropdown menu shows the user 'System Administrator' and the field changes '2026-09-20 04:41:14'. The record details are as follows:

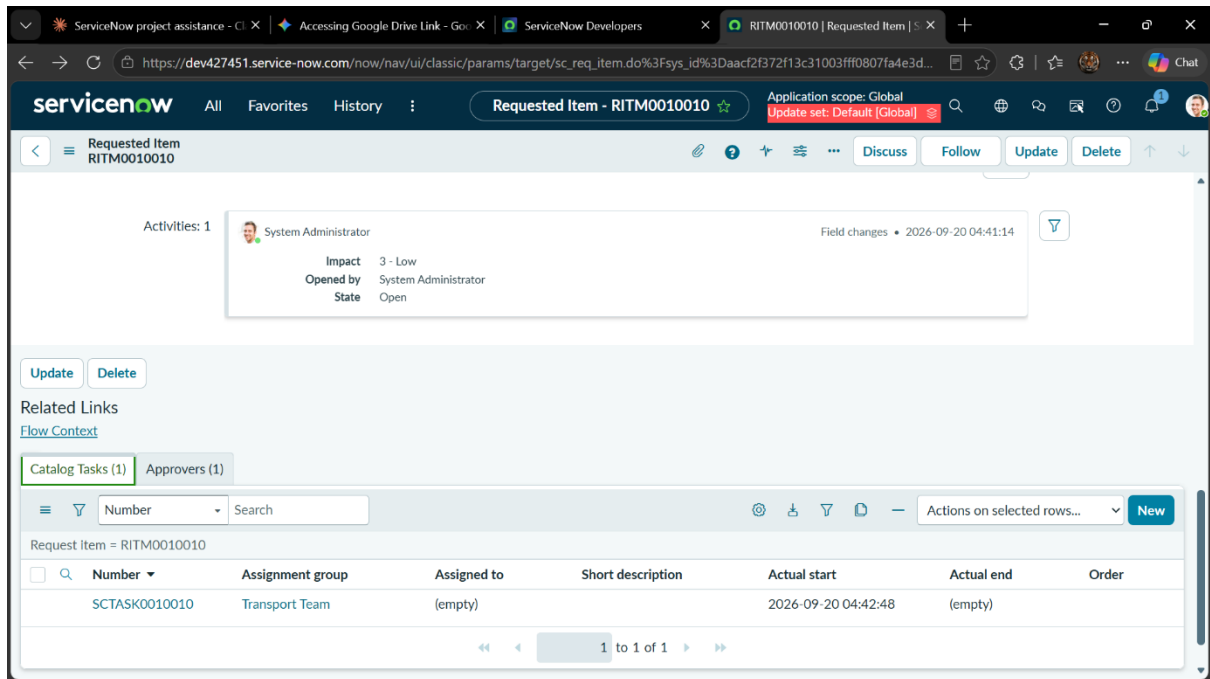
Field	Value
Impact	3 - Low
Opened by	System Administrator
State	Open

Buttons for 'Update' and 'Delete' are visible. Below the record details, the 'Related Links' section includes 'Flow Context'. The 'Catalog Tasks' tab is active, showing a table of approvers:

State	Approver	Comments	Created
Requested	Alejandra Prenatt		2026-09-20 04:41:23

The table shows 1 to 1 of 1 records.

Requested Item record with the approval state



Requested Item - RITM0010010

Activities: 1

System Administrator

Impact: 3 - Low
Opened by: System Administrator
State: Open

Field changes • 2026-09-20 04:41:14

Update Delete

Related Links

Flow Context

Catalog Tasks (1) Approvers (1)

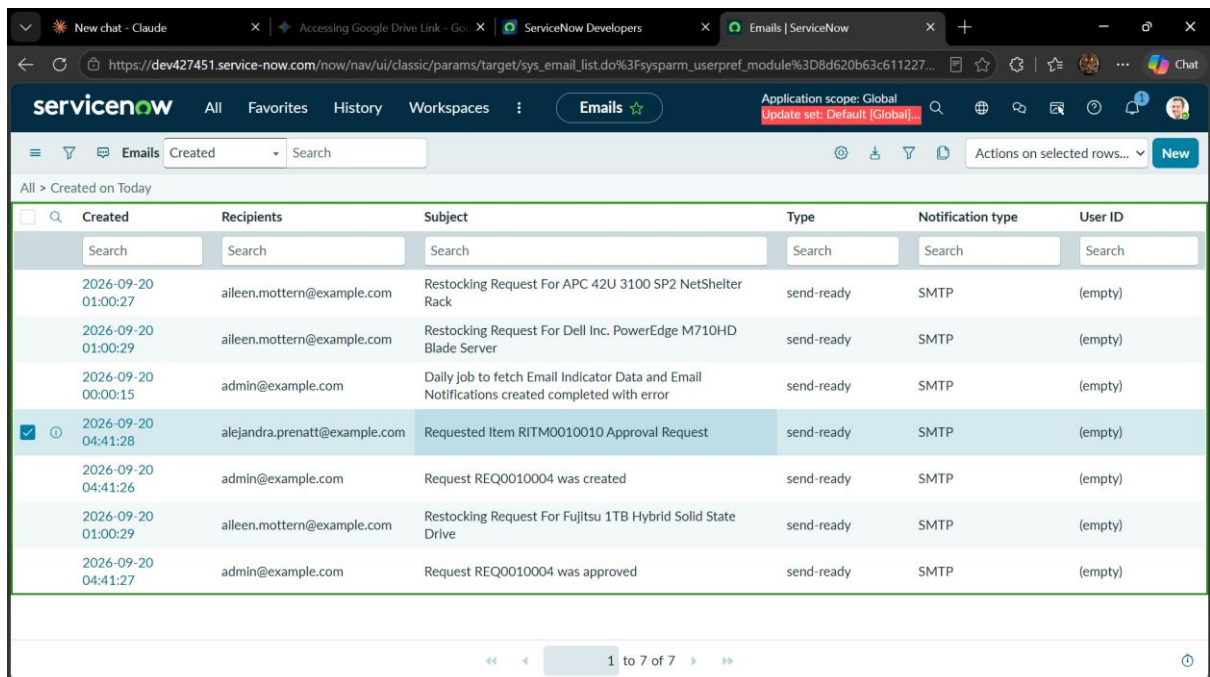
Number Search

Request Item = RITM0010010

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010010	Transport Team	(empty)		2026-09-20 04:42:48	(empty)	

1 to 1 of 1

Catalog task assigned to the Transport Team



Emails

Created Search

All > Created on Today

Created	Recipients	Subject	Type	Notification type	User ID
2026-09-20 01:00:27	aleen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-09-20 01:00:29	aleen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-09-20 00:00:15	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
2026-09-20 04:41:28	alejandra.prenatt@example.com	Requested Item RITM0010010 Approval Request	send-ready	SMTP	(empty)
2026-09-20 04:41:26	admin@example.com	Request REQ0010004 was created	send-ready	SMTP	(empty)
2026-09-20 01:00:29	aleen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-09-20 04:41:27	admin@example.com	Request REQ0010004 was approved	send-ready	SMTP	(empty)

1 to 7 of 7

Approval email in the email log or inbox

5.9 Finalize and Move the Update Set

1. Open the current Update Set and confirm it contains the catalog item, variables, choices, flow, and business rule.
2. Mark the Update Set as **Complete**.
3. Export it and load it on the target instance, preview it, and click **Commit**.

5.10 Troubleshooting Guide

Symptom	Likely cause	What to check
Flow never starts after ordering	Flow is not Active, or the trigger is not Service Catalog	Flow status, trigger type, and the flow's Executions list
No approval appears	Ask For Approval has no valid approver in the rule	The Rules section of the action
Task is not created	Approval State is not Approved, so the If branch is skipped	The If condition and the approval outcome
Email is not received	The task has not been completed (Wait), or the Requestor has no email	Task state, the user's email, and the email log
Incident appears unexpectedly	The business rule condition matches a state the request reaches normally	The value the rule checks and the state definitions
Changes are missing from the Update Set	A different Update Set was current while editing	The Update Set banner and the set's contents

5.11 Results

- A working end-to-end automation from request, to approval, to fulfilment task, to notification.
- An exception path that raises an incident through a Business Rule.
- All records are traceable through REQ, RITM, and task, with the approval history on the RITM.
- The configuration is packaged for deployment.

5.12 Limitations and Future Enhancements

Current behaviour	Enhancement
Only the approved path is automated	Add an Else branch that notifies the requester when a request is rejected
The approver is fixed in the rule	Use the requester's manager or an approval group dynamically
The email is sent only after the task is completed (Wait enabled)	Send an immediate approval email, and a second email on completion

Current behaviour	Enhancement
The email and task text do not mention the trip details	Add pickup, drop, date, duration, and car type to the task description and email body
Unavailability is handled after the fact by a Business Rule keyed on a state value	Add a Car Inventory table and check availability at request time; make the rule's condition explicit
Basic form validation	Add Catalog Client Scripts and mandatory fields, for example blocking past dates
No SLA tracking	Add an SLA definition on the RITM or the task

5.13 Conclusion

The project shows how ServiceNow's Service Catalog, Flow Designer, and Business Rules combine to replace a manual, email-driven car rental process with a structured and auditable workflow. Requesters get a single form, approvers receive requests automatically, and the Transport team gets a ready-to-act task. The design also leaves clear room to grow, from availability checks to SLA tracking.